INCH-POUND
MIL-DTL-24643/45F(SH)
w/INT. AMENDMENT 1
11 July 2011
USED IN LIEU OF
MIL-DTL-24643/45F
01 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS2U

This specification is approved for interim use by the Naval Sea Systems Command. Other activities in the Department of Defense may use this interim amendment or may continue using MIL-DTL-24643/45F.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, tin-coated, ASTM B286, 26 AWG – 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.015 inch: wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification code shall be applied by Method 3.
Fourth	-	Two conductors cabled together with a maximum lay of 1½ inches to form a pair.
Fifth	-	The specified number of pairs (see table I) shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Braided shield of tin-coated copper.
Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/45	Type and size	No. of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UO	LS2U-10	10	0.050	0.450	0.480
-02UO	LS2U-15	15	0.050	0.530	0.560
-03UO	LS2U-19	19	0.050	0.550	0.580
-04UO	LS2U-30	30	0.050	0.670	0.700
-05UO	LS2U-45	45	0.050	0.830	0.870
-06UO	LS2U-60	60	0.065	0.920	0.960

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	45.19
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Characteristic impedance, at 1 megahertz (MHz) (ohms)	135±15

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Bending endurance, At 25±2 °C (cycles, min.)	25

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Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (pounds per inch thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield (conformance to material, construction and coverage)	No failure
Surface transfer impedance (Type LS only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18

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Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2U – all sizes	1000

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Preparing Activity:
Navy – SH
(Project 6145-2011-024)

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